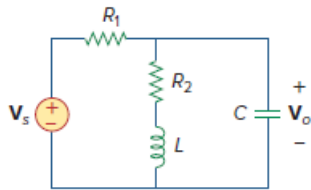


Problem

For the circuit in Fig. 10.71, determine V_o/V_s .

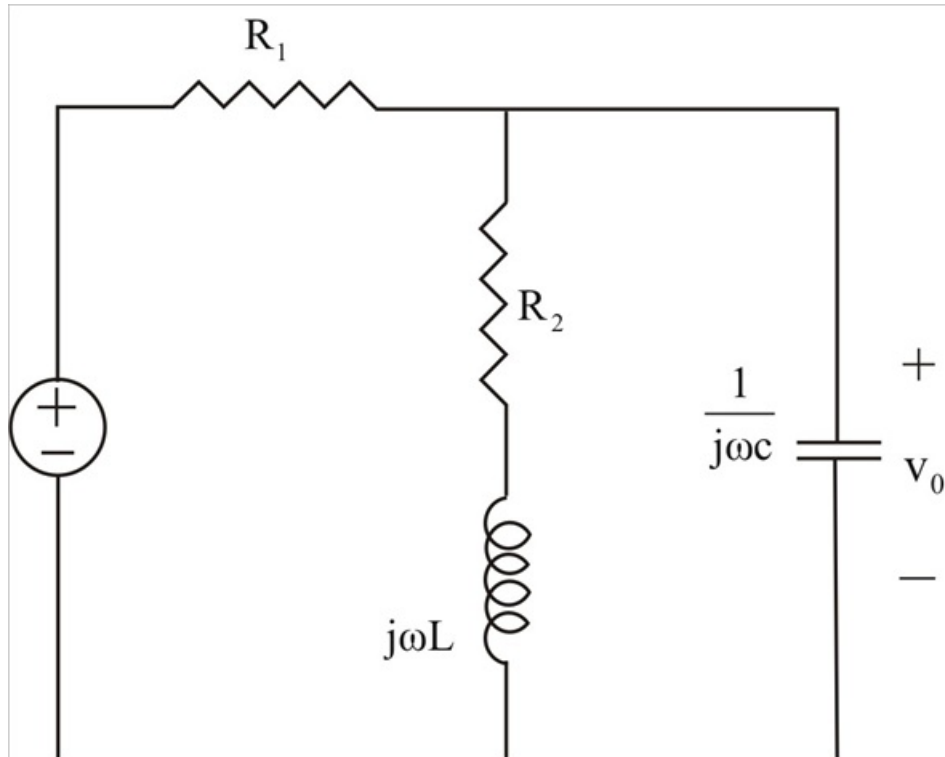
Figure 10.71:



Step-by-step solution

Step 1 of 2

Consider the circuit in the frequency domain as shown below.



Step 2 of 2

$$\text{Let } Z = (R_2 + j\omega L) \parallel \frac{1}{j\omega C}$$

$$Z = \frac{\frac{1}{j\omega C}(R_2 + j\omega L)}{R_2 + j\omega L + \frac{1}{j\omega C}} = \frac{R_2 + j\omega L}{1 + j\omega R_2 - \omega^2 LC}$$

$$\frac{V_0}{V_s} = \frac{Z}{Z + R_1} = \frac{\frac{R_2 + j\omega L}{1 - \omega^2 LC + j\omega R_2 C}}{R_1 + \frac{R_2 + j\omega L}{1 - \omega^2 LC + j\omega R_2 C}}$$

$$\frac{V_0}{V_s} = \frac{R_2 + j\omega L}{R_1 + R_2 - \omega^2 LC + j\omega(L + R_1 R_2 C)} \quad .$$